

EXPERIMENT 2

Aim

To implement and analyze Multiple Linear Regression, Ridge Regression, and Lasso Regression on a real-world house price dataset to study model performance, overfitting, and feature selection.

Problem Statement

House prices depend on multiple factors such as size, number of rooms, location, and amenities. The problem is to develop regression models that can:

- Predict house prices (continuous values) using Multiple Linear Regression
- Reduce overfitting using Ridge Regression (L2 Regularization)
- Perform feature selection using Lasso Regression (L1 Regularization)

The goal is to compare these models and identify the most efficient one for accurate prediction and better generalization.

Dataset Link

Link: <https://www.kaggle.com/code/chanakyavivekkapoor/house-price-prediction>

Methodology

1. Data Collection
 - Loaded the house price dataset from CSV file
2. Data Preprocessing
 - Removed columns with more than 40% missing values
 - Filled missing values using median (numerical) and "None" (categorical)
 - Converted categorical variables using one-hot encoding
3. Feature & Target Separation
 - Input Features (X): Property-related attributes
 - Target Variable (y): Price
4. Data Splitting
 - Split dataset into training and testing sets (80:20 ratio)
5. Feature Scaling
 - Applied StandardScaler to normalize features (important for Ridge & Lasso)
6. Model Training
 - a) Multiple Linear Regression
 - Trained baseline model to predict house prices
7. b) Ridge Regression
 - Applied L2 regularization to reduce overfitting

8. c) Lasso Regression
 - Applied L1 regularization for feature selection
9. Model Evaluation
 - Mean Squared Error (MSE)
 - R^2 Score
10. Hyperparameter Tuning
 - Used RidgeCV and LassoCV to find optimal alpha values
11. Feature Selection Analysis
 - Identified important features using Lasso (non-zero coefficients)

Screenshots:

(2000, 10)

	Id	Area	Bedrooms	Bathrooms	Floors	YearBuilt	Location	Condition	Garage	Price
0	1	1360	5	4	3	1970	Downtown	Excellent	No	149919
1	2	4272	5	4	3	1958	Downtown	Excellent	No	424998
2	3	3592	2	2	3	1938	Downtown	Good	No	266746
3	4	966	4	2	2	1902	Suburban	Fair	Yes	244020
4	5	4926	1	4	2	1975	Downtown	Fair	Yes	636056

(2000, 13)

Multiple Linear Regression
 MSE: 78279764120.86243
 R2 Score: -0.006181784611834162

Ridge Regression
 MSE: 78279030968.43915
 R2 Score: -0.006172360916937736

Lasso Regression
 MSE: 78279763978.74977
 R2 Score: -0.006181782785166012

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Total features: 13  
Selected features by Lasso: 13
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Best Ridge alpha: 50.0  
Best Lasso alpha: 0.01
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CONCLUSION

The results demonstrate that regularization techniques significantly improve regression models, especially for high-dimensional real-world datasets. Ridge Regression is preferable when all features are potentially useful, while Lasso Regression is ideal when feature selection is required.

Thus, Ridge and Lasso regression techniques offer better generalization and robustness compared to standard Multiple Linear Regression in complex datasets like house price prediction.